NTC 12.4.87

MILITARY SPECIFICATION

BEARING RESTORING BY RACEWAY GRINDING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for restoration of rolling element bearings.

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-C-320	Plating-Chromium (Electrodeposited) Class II

MILITARY

MIL-B-197	Bearing, Roller Element, Associated Parts
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-L-23699	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Aviation Research and Development Command, ATTN: DRDAV-ERS, St. Louis, MO 63120, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3110

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STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-271	Nondestructive Testing Requirements for Metals

(Copies of specifications, standards, drawings and publications required by Contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the effect on date invitation for bids or request for proposal shall apply.

AEROSPACE MATERIALS SPECIFICATIONS

AMS 2404	Plating-Nickel, Electroless
AMS 2410	Plating-Silver, Nickel Strike, High Bake
AMS 2412	Plating-Silver, Copper Strike, Low Bake
AMS 6419	Steel Bars, Forging and Tubing 4.1 CR-4.2 MO-1.0V (0.80-0.85C) Premium Bearing Quality, Double Vacuum Melted

(Application for copies should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendell, PA 15096.)

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI B46.1	Surface Texture

(Application for copies should be addressed to the American Standard Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM E 18	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM E 112	Average Grain Size of Metals
ASTM E 588-76	Detection of Large Inclusions in Bearing Quality Steel by the Ultrasonic Method

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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3. REQUIREMENTS

3.1 Inspection. All bearings components except rolling elements shall be visually inspected and separated into two categories: (1) nonrepairable and (2) repairable. Components in category (1) shall be rejected. All components in category (2) shall be nondestructive tested for the presence of cracks. The method shall be in conformance to MIL-STD-271. Components found acceptable shall further be dimensionally and hardness checked. Dimensional inspection shall be limited to mounting surfaces (ID, OD, width) and functional land surfaces. All plating shall be removed from separators prior to inspection. Inspection or test procedures used by the contractor shall be submitted and approved by the Government. Procedures shall not be changed without prior approval by the Government.

3.2 Rejection.

3.2.1 All bearings rolling elements shall be rejected without inspection (see 6.2.a).

3.2.2 Cages or separators not meeting original contractor's engineering drawing dimensions or specifications or exhibiting gross wear shall be rejected (see 6.2.b).

3.2.3 All races having nicks, dents, or other damage on the raceway surface extending to a depth greater than 0.002 inch or hardness not in conformance with the original contractor's engineering drawing shall be rejected.

3.2.4 Bearings which have been previously restored by grinding shall be rejected (see 3.6).

3.3 Rolling Element Material. New rolling elements shall be manufactured from materials which conforms to the original bearing manufacturer's drawing in two size categories as defined by para 3.4.3 herein. Material shall be produced by vacuum induction melting followed by vacuum consumable electrode remelting. This process used shall consistently produce ingots which yield billets free from voids and segregation. Inclusions shall not exceed the limits of AMS 6491 and ASTM E 588-76. The critical vacuum arc and other consumable electrode remelting process variable shall be continuously monitored and recorded if they exceed the established limits, the ingot shall not be used. Melting practices shall not be changed without Government approval. The crucible shall not have been used to melt materials containing undesirable constituents. Such materials may or may not be listed in the above specification. Example: Aluminum, magnesium, copper alloys, etc., have not been listed in the specification but shall be excluded (see 6.2.a and 6.5).

3.4 Process Requirements.

3.4.1 All raw materials shall be produced only from sources approved by the Government.

3.4.2 The bore surfaces of the inner rings, outside diameter surfaces of outer rings, face surface and land surfaces shall be restored to the original dimensions by chromium plating per QQ-C-320 or nickel plating per AMS 2404, if their pre-restoration condition does not meet their functional requirements.

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3.4.3 Raceway surfaces and roller flange surfaces on roller bearings shall be ground and finished by removing material from said surfaces to a depth as defined below:

The raceways shall be separated into two categories.

a. Those requiring only 0.003 inch (total inner plus outer) material removal to refurbish.

b. Those requiring 0.006 inch (total inner plus outer) material removal to refurbish.

Ths inner and outer raceways and guide flanges shall be ground to equal depths.

3.4.4 The correct number of new rolling elements shall be installed in the bearing.

3.4.4.1 New rolling elements installed in the bearing shall have a diameter equal to the original rolling element diameter plus twice the grinding depth in the two categories specified in 3.4.3, except for bearings with one race only, in which case the roller diameter shall be equal to the original roller diameter plus the grinding depth specified in 3.4.3 for that one race. The original print Internal Radial Clearance (IRC) shall be maintained.

3.4.4.2 Material for new rolling element shall conform to 3.3 and 3.5.

3.4.4.3 Where specified by the original bearing manufacturer, separators shall be stripped and replated in accordance with AMS 2410 or AMS 2412 as applicable. The separators shall be balanced to original specification after replating.

3.4.4.4 All manufacturing processes and procedures and inspection and quality control procedures used to produce parts of this specification shall be approved by the Government. Practices shall not be changed without approval by the Government.

3.5 Furnished Part Requirements.

3.5.1 Hardness. Rolling elements shall have an average hardness of Rockwell C 62-64. Rolling elements shall have a hardness within the range of Rockwell C 61-65. Race hardness shall be according to original bearing manufacturer's specification unless otherwise specified by the Government.

3.5.2 Dimensions and Tolerances. All dimensions and tolerances shall be in accordance with original manufacturer's engineering drawings and specifications except:

a. All rolling element dimensions shall be in accordance with 3.4.4.1.

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b. The surface finish of balls shall be 2 microinch AA and the rollers shall be 6 microinch AA or better on the O.D. and the ends. Raceway finish shall be in accordance with original specifications. Raceway surface finsih measurements shall be made in the axial direction.

3.5.3 Retained Austenite. The retained austenite of rolling elements shall not exceed 3 percent as measured in accordance with 4.3.

3.5.4 Grain Size. The average grain size of rolling elements shall be No. 8 or finer per ASTM E 112. Individual grains shall not exceed No. 5 per ASTM E 112 (see 4.2).

3.6 Traceability. Each restored bearing shall be assigned a number preceded by the letter "R" and the contractor's trademark or symbol. This number shall be marked on the face of each of the inner and outer rings. The bearing refurbishment contractor shall provide a record of traceability to the Government on the date of shipment of the bearing. The record shall show as a minimum the bearing serial number preceded by the letter "R", the contractor's trademark or symbol, the corresponding heat treat lot, the heat of steel, the bearing original part number, the date of shipment and the destination of shipment.

3.7 Material Record Requirements. Material identification record shall be prepared and furnished to the contracting officer on the day of final shipment. This record shall include, as a minimum, the following information:

a. Purchase order number.

b. Specific heat treatment cycle used (forger and/or bearing contractor).

c. Numerical results of all required tests and inspections.

d. Macro and micro examination test results.

e. Heat number.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Qualification Testing. A lot of ball and roller bearings restored in accordance with this specification shall be tested by the contractor, to establish the reliability of the bearings. A lot of bearings for qualification testing shall not be less than 30 bearings. Test conditions shall duplicate in a test rig those speeds, loads, and temperature conditions to which the bearing is subject in its applications. Bearing lubricant shall be MIL-L-23699 or MIL-L-7808. Test duration shall be the aircraft Time Between Overhaul (TBO) time or the bearing L10 life calculated in accordance with ASME methods, whichever is the shortest duration. Where the above procedure is not practical, accelerated load/life testing may be proposed. Test procedure and lubricant used shall be approved by the Government prior to initiation of this test. All bearings in a lot shall run to the time specified above without failure.

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4.1.2 Responsibility for Inspection. The contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to insure supplies and services conform to prescribed requirements. Government inspections shall not constitute acceptance; nor shall it replace the contractor's inspections; or otherwise relieve the contractor of his responsibility to furnish an acceptable end item.

4.1.3 Acceptance Inspection Test. All components and the assembled bearings shall be given a thorough mechanical and visual inspection and test to determine compliance with this specification. Specific details to be checked shall include, but not limited to, the following:

a. Identification markings and labels.

b. Dimensional accuracy.

c. Quality of finish.

d. Axial and radial looseness or "play".

e. Grinding burn.

f. Other visual defects.

4.1.4 Certified Test Report. The contractor shall furnish a certified test report showing that the contractor's product satisfactorily conforms to this specification. The test report shall include, as a minimum, actual results of the tests specified in 4.1.1. When the report is submitted, it shall be accomplished by a dated drawing which completely describes the contractor's processes of heat treat, finishes, part marking, and control specification. The contractor's part number for each size shall be included on the drawing.

4.1.5 Test After Delivery. The Government reserves the right to conduct any or all of the inspections or test specified herein, as deemed necessary after delivery of the bearing. Any defects discovered during or after these tests or inspections due to the contractor's failure to comply with drawings or specifications for his equipment shall be corrected by the contractor, at no expense to the Government.

4.2 Grain Size. Grain size shall be determined per ASTM E 112 on samples representing each heat treat lot of rolling element material. For reference tests, grain size shall be determined by a comparison of a polished and etched specimen with the chart in ASTM E 112 (see 3.5.4).

4.3 Retained Austenite. Retained austenite shall be determined by X-Ray diffraction techniques unless otherwise approved by the Government. If methods other than X-Ray diffractions are used, the method shall be calibrated to and show agreement with X-Ray diffraction measurement.

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4.4 Hardness. Hardness tests shall be conducted in accordance with ASTM E 18.

4.5 Surface Finish. Surface finish shall be determined in accordance with ANSI B46.1.

5. PACKAGING

5.1 Bearings shall be packaged, packed and marked Level A, B, or C, as specified (see 6.3) in accordance with MIL-B-197. In addition to any special marking specified in the contract order, containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use. Restoration of Government owned rolling element bearings by a grinding process.

6.2 Definitions. For purposes of this specification, the following definitions shall apply:

a. Rolling Element Bearings. A rolling element bearing is a bearing having an inner race, an outer race, and a plurality of either balls, cylindrical rollers, crowned rollers, or spherical rollers.

b. Cage or Separator. A cage, or separator, separates positions and/or retains the rolling elements between the inner and the outer raceways.

c. Restoration. The process of grinding raceways, plating (diameters, faces, lands, and cages) and replacing rolling elements in accordance with this specification.

6.3 Ordering Data. Procurement document should specify the following:

a. Level of preservation, packaging and packing required (see 5.1).

b. Whether special marking for shipment and storage is required (see 5.1).

6.4 Classification of Characteristics.

a. Critical (see paragraphs 3.2.3, 3.4.3, 3.4.4.1, 3.4.4.3 and 3.5).

b. Major (see paragraphs 3.5.2 and 3.5.3).

c. Minor (all other paragraphs).

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

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